

Towards a semi-empirical model of the sea ice thickness based on hyperspectral remote sensing in the Bohai Sea

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Abstract

Sea ice thickness is one of the most important input parameters for the prevention and mitigation of sea ice disasters and the prediction of local sea environments and climates. Estimating the sea ice thickness is currently the most important issue in the study of sea ice remote sensing. With the Bohai Sea as the study area, a semi-empirical model of the sea ice thickness (SEMSIT) that can be used to estimate the thickness of first-year ice based on existing water depth estimation models and hyperspectral remote sensing data according to an optical radiative transfer process in sea ice is proposed. In the model, the absorption and scattering properties of sea ice in different bands (spectral dimension information) are utilized. An integrated attenuation coefficient at the pixel level is estimated using the height of the reflectance peak at 1 088 nm. In addition, the surface reflectance of sea ice at the pixel level is estimated using the 1 550–1 750 nm band reflectance. The model is used to estimate the sea ice thickness with Hyperion images. The first validation results suggest that the proposed model and parameterization scheme can effectively reduce the estimation error associated with the sea ice thickness that is caused by temporal and spatial heterogeneities in the integrated attenuation coefficient and sea ice surface. A practical semi-empirical model and parameterization scheme that may be feasible for the sea ice thickness estimation using hyperspectral remote sensing data are potentially provided.

Key words: Bohai Sea, sea ice thickness, hyperspectral remote sensing, semi-empirical model

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1 Introduction

The area around the Bohai Sea is the most important economic area in northern China. The total marine production value in the area reached RMB 2.215 2×10¹⁰ Yuan in 2014, accounting for 37.0% of the gross national marine production value (State Oceanic Administration, 2015). In the Bohai Sea, sea ice seriously threatens marine traffic safety, marine aquaculture, and offshore oil platforms during the winter (Li et al., 2004; Gu et al., 2013). Via Chinese sea ice desalinization projects, sea ice (brackish water) can be converted into fresh water, providing new fresh water resources for socioeconomic development in areas along the coast of the Bohai Sea (Gu et al., 2012). Understanding the sea ice thickness in the Bohai Sea is of great significance to the exploration of local air-sea heat exchange mechanisms (Xie et al., 2001), sea ice engineering design (Yang, 2000), prevention and mitigation of sea ice disasters, and sea ice resource utilization (Guo et al., 2008; Liu et al., 2013).

Compared with the sea ice thicknesses of several metres or even dozens of metres near the South and North Poles, sea ice in the Bohai Sea is relatively thin. In the Bohai Sea, no polar night phenomenon occurs during the winter, and weather conditions are relatively good. In general, the amount of winter snowfall in

the Bohai Sea is small. Because of the wind and current, a snow cover is rare on the sea ice (Yang, 2000). If snowfall does occur, typically only a few centimetres accumulate (Yang, 2000). Therefore, optical remote sensing images, thermal infrared remote sensing images, and microwave remote sensing data are all used to determine or estimate the area and thickness of sea ice in the Bohai Sea.

In terms of sea ice monitoring by optical remote sensing in the Bohai Sea, based on a relation between an albedo and the sea ice thickness, Xie et al. (2003) first proposed an exponential model (EM) between the albedo and the sea ice thickness and estimated the sea ice thickness using advanced very high resolution radiometer (AVHRR) images. Luo et al. (2004) proposed an empirical piecewise linear model based on Haiyang-1 (HY-1) images. Wu et al. (2006) used the empirical piecewise linear model and moderate resolution imaging spectroradiometer (MODIS) Band 1 and Band 8 images to estimate the sea ice thickness in the Bohai Sea. Ning et al. (2009) used thematic mapper (TM) Band 2 or MODIS Band 4 data to distinguish sea ice types and determine the distribution of the sea ice thickness based on the relationship between the sea ice type and ice thickness. Yuan et al. (2012) validated the EM model using the AVHRR images and the *in situ*

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measured ice thickness. Their results indicate that the average error of inversion associated with the sea ice thickness is below 30%. Su and Wang (2012) used the MODIS images to estimate the sea ice thickness in the Bohai Sea using the EM model, finding that the estimated sea ice thicknesses were in accordance with the Lebedev and Zubov empirical model results and the forecasting data from the National Marine Environmental Forecasting Centre of China (NMEFC), the State Oceanic Administration. In terms of sea ice monitoring by thermal infrared remote sensing, Wu et al. (2005) proposed an empirical model based on MODIS brightness temperature data and sea ice thickness.

In terms of sea ice monitoring by microwave remote sensing, Zhang (2011) suggested that microwaves, including the X-band, C-band, and L-band, can penetrate a certain thickness of sea ice, and the order of the penetration depth is L-band greater than C-band greater than X-band. Zhang (2011) also proposed a sea ice thickness level detection method based on a binary-tree classification theory and C-band synthetic aperture radar (SAR) data. Zhang et al. (2013) combined a multilayered sea ice electromagnetic scattering model with a sea ice thermodynamic model to determine the thickness of flat, thin ice using SAR data at different frequencies, polarizations, and incidence angles. The results indicate that copolarization backscattering coefficients and the phase of co-polarized correlation are correlated with the sea ice thickness positively, while the co-polarized ratio and the magnitude of the co-polarized correlation (including for L-band SAR) are negatively related to the sea ice thickness. Zhang et al. (2015) proposed a three-component (surface, double-bounce, volume, and residual components) scattering model to decompose polarimetric SAR (PolSAR) sea ice data. The validation results of the proposed model based on the C-band Radarsat-2 quad-polarization data acquired over sea ice in the Bohai Sea show that the proposed model helps to distinguish different ice types and offers a proxy for the sea ice thickness. Liu et al. (2015b) proposed a principal component analysis-based (PCA-based) sea ice image fusion of optical data using HIS transform and SAR data using wavelet transform. Additionally, this fusion method can improve an image interpretation and a sea-ice identification through objective, subjective, and comprehensive evaluations.

Moreover, there some hybrid methods also exist. Shi et al. (2014) presented an sea ice thickness estimation method that applied a high-resolution thermodynamic snow and ice model (HIGHTSI) to calculate a thermodynamic ice growth, which was used as the sea ice thickness background. SAR data were used to express the local ice statistics and to redistribute the modelled ice thickness. The method was validated with *in situ* observations from oil platforms and ice forecast results from NMEFC. Liu et al. (2015a) established an empirical logarithmic model to estimate level ice thickness from visible/near-infrared (VIS/NIR) imagery from Huanjing-1 (HJ-1) and an empirical exponential model to estimate rough ice thickness based on ENVISAT SAR data.

In general, the models and the methods are used to estimate the sea ice thickness in the Bohai Sea based on optical, thermal infrared, and microwave remote sensing data are relatively qualitative and empirical. Some parameters must generally be determined at regional or global scales based on experience. These parameters may be influenced by spatial and temporal variations associated with sea ice properties, resulting in large errors. Thus, to improve the sea ice thickness estimation accuracy in the Bohai Sea, it is necessary to develop a model or method with parameters that can be determined at the pixel scale from remote sensing data or other actual measured data.

In this paper, we simplified the complex sea ice radiative transfer model and proposed a semi-empirical model of the sea ice thickness (SEMSIT) with hyperspectral remote sensing images as the data source. To evaluate the SEMSIT, a hyperspectral remote sensing image obtained by Hyperion mounted on the earth observing-1 (EO-1) satellite on January 21, 2008 was used to estimate the sea ice thickness. The EO-1 satellite was launched in November 2000 by the United States. It has an orbital altitude of 705 km and follows a sun-synchronous orbit (Pearlman et al., 2003). Hyperion is an imaging spectrometer with 220 bands from 357 nm to 2 576 nm (Beck, 2003). The spectral resolution of Hyperion is approximately 10 nm, and the spatial resolution is 30 m (Pearlman et al., 2003). Based on preliminary estimation results, the proposed model can greatly reduce the estimation error associated with the sea ice thickness caused by temporal and spatial heterogeneity in the integrated attenuation coefficient and surface reflectance of sea ice, and improve the accuracy of the sea ice thickness estimates based on the optical remote sensing images compared with the exponential model adopted by Xie et al. (2003), Yuan et al. (2012), and Su and Wang (2012). On the basis of the proposed model, hyperspectral remote sensing technology may be effective for monitoring sea ice in the Bohai Sea.

2 Study area and data

2.1 Study area

The Bohai Sea geographically extends from 37°07'N to 41°00'N and from 117°35'E to 122°15'E (Fig. 1). As a nearly closed shallow sea extruding into Chinese Mainland, the Bohai Sea is roughly triangular. Three protruding corners of the triangle correspond to the Liaodong Bay, the Bohai Bay, and the Laizhou Bay (Feng et al., 1999). In the Bohai Sea, sea ice is first-year ice, and the ice period lasts from late November to March of the following year. The sea ice thickness across the majority of the Bohai Sea usually varies from several centimetres to tens of centimetres, except sea ice that accumulates near the coast, which may be several metres thick (Ji and Yue, 2011).

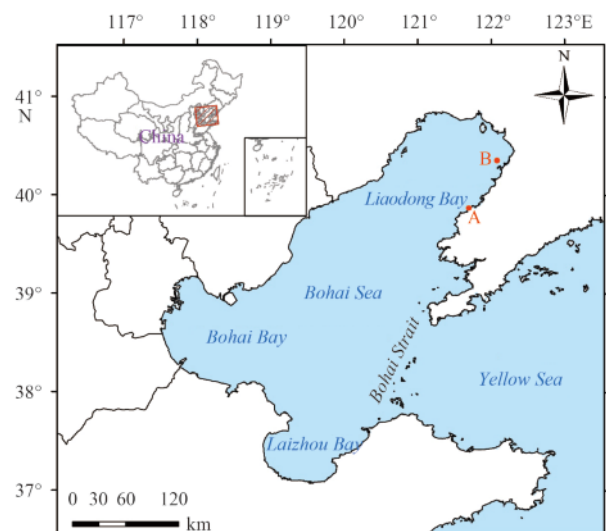


Fig. 1. Overview of the Bohai Sea. Points A and B are the field measurement sites where the thicknesses and reflectance spectra of sea ice were collected in the winters of 2011 to 2012 and 2012 to 2013.

2.2 Measured sea ice reflectance spectra

In situ sea ice reflectance values were measured based on 37 ice samples from Points A and B shown in Fig. 1 during the winters of 2011–2012 and 2012–2013. The measured reflectance values of sea ice are shown in Fig. 2. Moreover, the reflectance of new snow was also measured in the winter of 2011–2012 (Fig. 3). To make the samples representative, the reflectance spectra of ice samples with different thicknesses (from 5 cm to 55 cm) were collected. The ice thicknesses are shown in Table 1. The weather was sunny during the sampling period. The sea ice reflectance spectra at each sample point were collected using a FieldSpec 3 spectroradiometer made by analytical spectral devices, Inc., United States. The spectroradiometer covers the full range (350–2 500 nm), which is the concentration range associated with solar radiation energy and the imaging wavelength range of most optical satellites. It outputs the spectra with a sample interval of 1 nm.

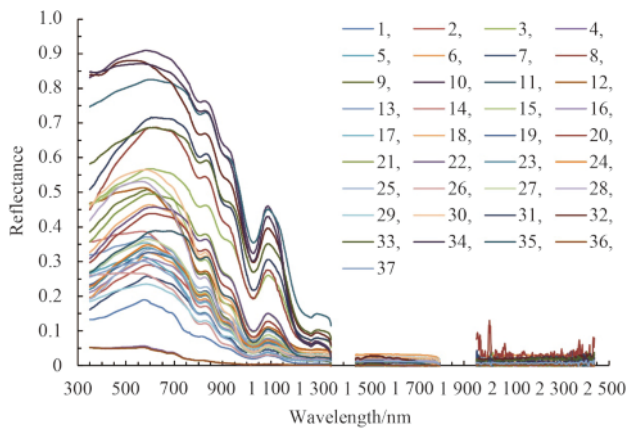


Fig. 2. The measured reflectance values of sea ice with different thicknesses (cm). The reflectance spectra were measured at Points A and B shown in Fig. 1 in the winters of 2011–2012 and 2012–2013. The legend denotes the IDs of ice samples.

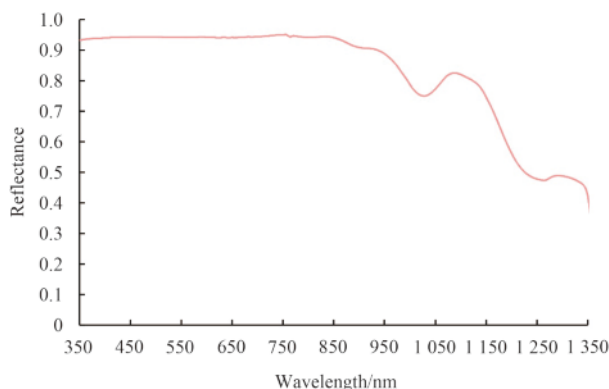


Fig. 3. Measured reflectance of new snow.

2.3 Remote sensing data

Hyperion is mounted on the earth observation satellite EO-1, which was launched in November 2000 in the United States (Pearlman et al., 2003). The orbital altitude of the EO-1 satellite is 705 km. It follows a sun-synchronous orbit, with an inclination angle of 98.2°. The working band of the Hyperion sensor is 357–2 576 nm and is divided into 220 subbands (Beck, 2003). The spa-

tial resolution is 30 m, and the spectral resolution is approximately 10 nm (Pearlman et al., 2003). Because the satellite is experimental, the swath width of Hyperion images is limited to 7.7 km. Although its coverage is small, it has been widely used in numerous studies and experiments related to agriculture, forestry, mining, oceanography, and other fields.

The adopted Hyperion images were acquired over the western Bohai Sea at 02:19:15 am UTC on January 21, 2008. The solar elevation angle during image acquisition was 24.11°, and the sun azimuth was 151.53°. The viewing zenith angle of the sensor was 11.43°. The images are of level L1Gst, as obtained after a geometric correction, a projection registration, and a topographic correction (Fig. 4). The L1Gst product has 242 bands, but only 198 bands are calibrated, as shown in Table 2 (Beck, 2003). Table 2 summarizes the Hyperion spectral coverages and band designators.

3 Model

In this section, according to the optical radiative transfer process in sea ice, the proposed semi-empirical model of the sea ice thickness is described. The assumptions, the determination of the parameters in the model and the advantages are also detailed.

3.1 The optical radiative transfer process in sea ice

Figure 5 shows the general radiative transfer process of the solar radiation in sea ice. First, after passing through the atmosphere, the solar radiation arrives on the sea ice surface. A small portion of the radiation is reflected back to the atmosphere, and the remainder of the radiation enters into the sea ice. Second, via scattering and absorption by ice crystals, brine pockets, and bubbles, a portion reenters the atmosphere, and the other portion reaches the sea ice bottom. Third, a small portion of the radiation that arrives on the sea ice bottom is reflected by sea water, passing through the sea ice and reentering the atmosphere. For thicker ice, the total sea ice attenuation is so large that the solar radiation may be unable to penetrate to the ice bottom. Therefore, whether the solar radiation can penetrate to the ice bottom or not depends on the ice thickness and the wavelength (sea ice may have different absorption and scattering properties at different wavelengths). Thus, the radiative transfer process may only involve the first two steps at a certain wavelength for some sea ice conditions. Currently, the radiative transfer equation can accurately describe the optical radiative transfer process in sea ice (Grenfell, 1983; Light et al., 2003; Zege et al., 2013); however, it requires numerous auxiliary parameters. Moreover, the calculation process is complex and not suitable for the inversion process associated with the sea ice thickness based on remote sensing images.

Therefore, it is necessary to simplify the radiation transfer equation, highlight the main issues, and obtain a practical semi-empirical model. In fact, many remote sensing models have been obtained in this manner. According to different sources, the radiation L , $W/(m^2 \cdot \mu m \cdot sr)$ received by the sensor above sea ice is associated with ① sea ice surface scattering (S), ② sea ice volume scattering C entering the atmosphere, and ③ sea ice bottom reflection B entering the atmosphere. The sea ice volume scattering C entering the atmosphere and bottom reflection B entering the atmosphere occur within sea ice and can be denoted as internal scattering (D , where $D=B+C$) from sea ice. The radiation received by the sensor above sea ice can be expressed as follows:

$$L = S + D. \quad (1)$$

Table 1. The thicknesses of ice samples

ID ¹⁾	Thickness/cm	ID	Thickness/cm	ID	Thickness/cm	ID	Thickness/cm
1	9.5	11	41.1	20	29.5	29	28.7
2	32.5	12	35.3	21	32.2	30	55.0
3	30.1	13	32.2	22	30.0	31	33.2
4	5.5	14	31.2	23	35.0	32	39.4
5	35.1	15	41.5	24	35.2	33	32.5
6	30.6	16	30.1	25	28.2	34	37.8
7	30.1	17	32.2	26	30.0	35	30.1
8	36.2	18	34.1	27	34.0	36	5.2
9	43.8	19	29.0	28	41.8	37	31.9
10	42.5						

Note: ¹⁾ The IDs correspond to ice samples in Fig. 1.

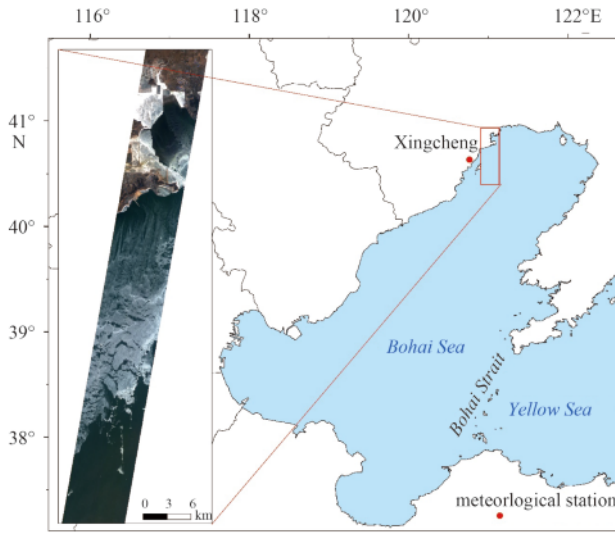


Fig. 4. The Hyperion image (acquired on January 21, 2008). The red solid circle shows the location of the nearest meteorological station.

Table 2. Summary of Hyperion spectral coverages and band designators

	Band	Wavelength/nm	Calibrated	Used in SEMSIT
VIS/NIR	1–7	355.6–416.6	no	no
	8–29	426.8–640.5	yes	no
	30–33	650.7–681.2	yes	yes
	34–70	650.7–1057.7	no	no
SWIR	71–76	851.9–902.4	no	no
	77–84	912.5–983.1	yes	no
	85–114	993.2–1 285.8	yes	yes
	115–139	1 295.9–1 537.9	yes	no
	140–160	1 548.0–1 749.8	yes	yes
	161–224	1 759.9–2 395.5	yes	no
	225–242	2 405.6–2 577.1	no	no

3.2 Semi-empirical model of sea ice thickness

According to the optical radiative transfer process in sea ice noted above, the sea ice reflectance for a certain wavelength can be expressed as follows:

$$\rho = R^S + R^D, \quad (2)$$

where R^S refers to the sea ice surface reflection, corresponding to

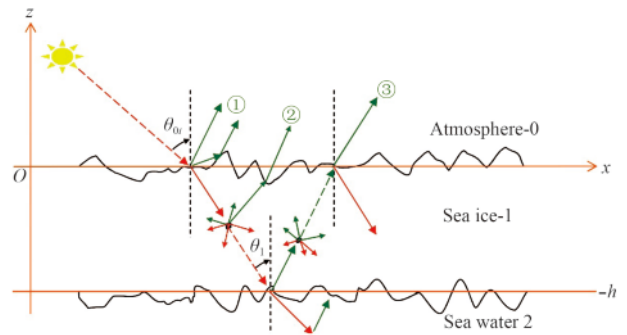


Fig. 5. Schematic diagram of the solar radiation transfer process in sea ice. ① Sea ice surface scattering. ② Sea ice volume scattering entering the atmosphere. ③ Sea ice bottom reflection entering the atmosphere. θ_{0i} is the sun zenith angle. θ_1 is the zenith angle of the radiation incident at the ice-water interface.

the first item on the right side of Eq. (1); and R^D refers to the internal scattering and reflection entering the atmosphere, corresponding to the second item on the right side of Eq. (1). The radiative transfer process of the solar radiation in sea ice is similar to that in sea water. Therefore, based on the model for a water body reflectance by Lee et al. (1999), R^D can be expressed as follows:

$$R^D = \frac{\alpha r}{1 - \beta r}, \quad (3)$$

where r is the reflectance below the sea ice-air interface; α and β are coefficients. The reflectance below the sea ice-air interface (Lee et al., 1999) can be expressed as follows:

$$r = r_\infty [1 - e^{-(k_i + k_c)h}] + r_w e^{-(k_i + k_b)h}, \quad (4)$$

where r_∞ is the subsurface reflectance of optically thick sea ice; r_w is the reflectance of sea water below sea ice; k_i is the downward extinction coefficient, k_c is the total upward extinction coefficient; k_b is the extinction coefficient between the sea ice-water interface and sea ice-air interface; and h is the sea ice thickness, where $h \leq H_0$. H_0 is the penetration depth of the solar radiation in sea ice and is closely related to the sea ice extinction coefficient. When sea ice thickness increases above H_0 , the sea ice reflectance reaches a maximum value, becoming insensitive to increases in the sea ice thickness.

There are too many unknown parameters (R^S , α , β , r_∞ , r_w , k_i , k_b , k_c) in Eqs (2)–(4) to estimate the sea ice thickness directly using Eqs (3) and (4) based on hyperspectral or multispectral re-

remote sensing data. In addition, these parameters generally vary based on the spatial and temporal variations of sea ice properties, which are major error sources. To improve the accuracy of sea ice estimation and establish a practical estimation model, it is necessary to determine these parameters at the pixel scale. In this study, we adopt such a strategy, which reasonably simplifies the model described by Eqs (2)–(4) and determines the parameters based on the practical and empirical relationships between the parameters and hyperspectral remote sensing data or other actual measured data.

First, we merge and reduce the parameters in Eqs (2), (3), and (4) to determine the parameters based on the hyperspectral remote sensing data or other actual measured data. Lee et al. (1999) published the empirical calculation methods for k_b and k_c for sea water. These coefficients can be obtained through Hydrolight software simulations (Mobley, 1995) with different absorption coefficients, different scattering coefficients, and additional input parameters. Owing to the lack of available coefficients associated with sea ice in the Bohai Sea and simulation software limitations, it is difficult to calculate k_b and k_c for sea ice. In Lee et al. (1999), k_b and k_c have the same functional form. We also assume that the functional forms of k_b and k_c are similar because the radiative transfer process of the solar radiation in sea ice is similar to that in sea water. Therefore, we assumed that $k_c \approx k_b$ in Eq. (4) rather than focusing on concrete calculation methods. Although the assumption may result in a certain error, it helps to calculate h using Eq. (4) based on the hyperspectral remote sensing data. On the basis of the above simplification, Eq. (4) can be rewritten as follows:

$$r = r_\infty [1 - e^{-\mu h}] + r_w e^{-\mu h}, \quad (5a)$$

$$h = -\frac{1}{\mu} \ln \frac{r_\infty - r}{r_\infty - r_w}, \quad (5b)$$

$$\mu = -\frac{1}{h} \ln \frac{r_\infty - r}{r_\infty - r_w}, \quad (5c)$$

where μ is the integrated attenuation coefficient of sea ice given by $\mu = k_a + k_c \approx k_a + k_b$. Here, μ is not the complete extinction coefficient. It expresses more information about the integrated weakening effects of sea ice associated with the solar radiation.

Lee et al. (1999) also obtained the coefficients in Eq. (3) for sea water using Hydrolight software simulations. The coefficients in Eq. (3) are related to the refractive index and surface roughness of sea ice/sea water. Because a sea ice surface morphology is quite different from that of the seawater surface and the reflectance below the sea ice surface is much larger than that below the seawater surface, the coefficients obtained by Lee et al. (1999) in Eq. (3) are not suitable for sea ice. For sea ice in the Bohai Sea, there is also no precise estimation model for the sea ice surface morphology, excluding a few field observations obtained from the Bohai Sea (Li et al., 2009; Liu et al., 2014b). Assuming that the air-ice and ice-water transmittances are $t_{a,i}$ and $t_{i,w}$, respectively, and the subsurface reflectance of sea water below sea ice is r_{sb-w} , we assumed that $R^D \approx t_{a,i} r$ and $r_w \approx t_{i,w} r_{sb-w}$. After multiplying both sides by the air-ice transmittance, Eq. (5a) can be rewritten as follows:

$$R^D = t_{a,i} r_\infty [1 - e^{-\mu h}] + t_{a,i} t_{i,w} r_{sb-w} e^{-\mu h}, \quad (6)$$

where $t_{a,i} r_\infty$ can be considered the supersurface reflectance of optically thick sea ice. Because the refractive index of ice is very

close to that of sea water (Warren, 1984; Austin and Halikas, 1976), we assume that $t_{i,w} \approx 1.0$, the air-ice transmittance is equal to the air-water transmittance, and $t_{a,i} t_{i,w} r_{sb-w} \approx t_{a,i} r_{sb-w}$. Thus, $t_{a,i} r_{sb-w}$ also can be considered the supersurface reflectance of seawater. Combined with Eq. (2), Eq. (6) can be rewritten as follows:

$$\rho = \rho_\infty [1 - e^{-\mu h}] + \rho_w e^{-\mu h} + R^S, \quad (7a)$$

$$h = -\frac{1}{\mu} \ln \frac{\rho_\infty - \rho + R^S}{\rho_\infty - \rho_w}, \quad (7b)$$

$$\mu = -\frac{1}{h} \ln \frac{\rho_\infty - \rho + R^S}{\rho_\infty - \rho_w}, \quad (7c)$$

where ρ_∞ is the supersurface reflectance of optically thick sea ice; ρ_w is the supersurface reflectance of sea water, which is the reflectance of sea ice with a thickness of 0; and μ indicates the integrated weakening effects of sea ice on the solar radiation, including the volume scattering and volume absorption of sea ice. As long as R^S , ρ_∞ , ρ_w and μ are precisely estimated, the sea ice thickness h can be accurately estimated. In this manner, the semi-empirical model of the sea ice thickness was established.

3.3 Determination of the parameters

Equations (7a)–(7c) are theoretically applicable to any band in the reflection spectrum. However, the sea ice thickness h cannot be estimated using the reflectance of only one band unless four additional parameters are assumed in advance. Thus, for all relevant pixels, the four parameters in Eqs (7a)–(7c) remain unchanged. As mentioned above, due to the spatial and temporal differences in the scattering coefficients and absorption coefficients of sea ice, holding these four parameters constant for all of the pixels will lead to large errors, especially errors caused by μ and R^S . Therefore, we estimated the four parameters based on the correlation of the hyperspectral remote sensing data in spectral-dimension or spatial-dimension.

3.3.1 Band selection for the estimation of sea ice thickness

To estimate h , we must select a band with a large sea ice penetration depth that is sensitive to sea ice thickness variations. Figure 6 shows the real and imaginary parts of the complex refractive index of pure ice within the 400–2 500 nm band. The data are

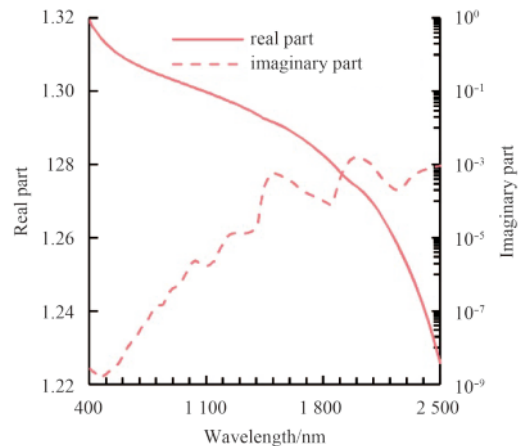


Fig. 6. The real and imaginary parts of the complex refractive index of pure ice.

from the results of Warren (1984). The real part of the refractive index of pure ice exhibits no significant difference within the band of 400–2 500 nm, while the imaginary part exhibits a large difference within the band of 400–2 500 nm, increasing with wavelength. Figure 7 shows the absorption coefficient and penetration depth calculated based on the imaginary part of the refractive index. The maximum penetration depth is approximately 24.13 m, occurring at 470 nm. Within the band of 400–1 430 nm, the penetration depth gradually increases and reaches a maximum depth at a wavelength of 470 nm. The greater penetration depth leads to more radiation energy, which enters sea ice and is scattered by brine pockets and bubbles, and a larger reflectance. Therefore, the reflectance at 470 nm is the theoretical maximum reflectance; however, the maximum reflectance shown in Fig. 7 is not measured at 470 nm, but instead at 618 nm. The difference can be interpreted as follows. As wavelength decreases, the scattering capability of particles exponentially increases. However, multiple scattering among particles will weaken the total backscattering capacity. Therefore, a shorter wavelength represents a slower increase in the reflectance. Thus, we chose the 650–690 nm band to estimate the sea ice thickness h . According to our parameterization scheme, Eqs (7a)–(7c) can be rewritten as follows:

$$\rho_{650-690} = \rho_{\infty, 650-690} [1 - e^{-\mu_{650-690}h}] + \rho_{w, 650-690} e^{-\mu_{650-690}h} + R_{650-690}^S, \quad (8a)$$

$$h = -\frac{1}{\mu_{650-690}} \times \ln \frac{\rho_{\infty, 650-690} - \rho_{650-690} + R_{650-690}^S}{\rho_{\infty, 650-690} - \rho_{w, 650-690}}, \quad (8b)$$

$$\mu_{650-690} = -\frac{1}{h} \ln \frac{\rho_{\infty, 650-690} - \rho_{650-690} + R_{650-690}^S}{\rho_{\infty, 650-690} - \rho_{w, 650-690}}, \quad (8c)$$

where $\rho_{650-690}$ is the sea ice reflectance within the band of 650–690 nm; $\rho_{\infty, 650-690}$ is the supersurface reflectance of optically thick sea ice within the band of 650–690 nm; $\rho_{w, 650-690}$ is the supersurface reflectance of sea water within the band of 650–690 nm; $\mu_{650-690}$ is the integrated weakening effects of sea ice on the solar radiation within the band of 650–690 nm; and $R_{650-690}^S$ is the sea ice surface reflectance within the band of 650–690 nm.

3.3.2 Determination of $\rho_{\infty, 650-690}$ and $\rho_{w, 650-690}$

$\rho_{\infty, 650-690}$ and $\rho_{w, 650-690}$ correspond to the reflectance of sea ice

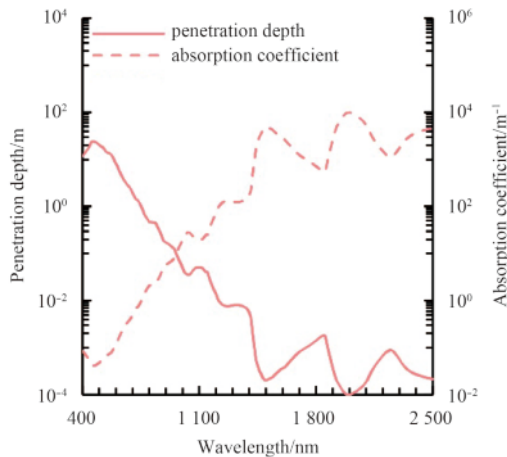


Fig. 7. The absorption coefficient of pure ice.

with a thickness that is larger than the penetration depth and the reflectance of sea water, respectively. Lee et al. (1999) established an empirical model to calculate the reflectance for optically deep water. The model input is the single scattering albedo of seawater. For sea ice in the Bohai Sea, no model is available to calculate $\rho_{\infty, 650-690}$ using the single scattering albedo or to estimate $\rho_{w, 650-690}$ from the remote sensing data at the pixel scale. Additionally, no model is available to calculate the single scattering albedo based on remote sensing data at the pixel scale. Similar to Xie et al. (2003), Yuan et al. (2012), and Su and Wang (2012), we treated $\rho_{\infty, 650-690}$ as a fixed value. We assumed that the reflection of snow is the limitation of the reflection of sea ice with a thickness larger than the penetration depth. In this study, we set $\rho_{\infty, 650-690}$ as the reflectance of snow within the band of 650–690 nm, a value of 0.94 (Fig. 3). $\rho_{w, 650-690}$ in Eq. (8b) was estimated from the adjacent ice-free pixels using the inverse distance weighted interpolation algorithm.

3.3.3 Determination of $R_{650-690}^S$ and $\mu_{650-690}$

To determine $R_{650-690}^S$, the selected band should only contain (or be dominated by) sea ice surface reflection. To determine $\mu_{650-690}$, the selected band should reflect the internal information associated with sea ice and not be sensitive to sea ice thickness variations (moderate penetration depth). According to Figs 5 and 6, when the wavelength is greater than 1 430 nm, the penetration depth is less than 0.002 m. The radiation energy of sea ice within the band of 1 430–2 500 nm is mainly due to surface reflection and not from internal sea ice. Therefore, the sea ice reflectance is close to or equal to the sea ice surface reflectance within the band of 1 430–2 500 nm. Within the band of 400–1 430 nm, the penetration depth decreases with wavelength. Thus, we chose the 1 550–1 750 nm band to estimate $R_{650-690}^S$. $R_{650-690}^S$ can be estimated as follows:

$$R_{650-690}^S = R_{1 550-1 750}, \quad (9)$$

where $R_{1 550-1 750}$ is the reflectance of the 1 550–1 750 nm band.

$\mu_{650-690}$ is related to the internal structure of sea ice. According to previous results by Liu et al. (2014b), the sea ice density is negatively correlated with the sea ice density spectrum index (SIDS) which is the height of reflectance peak at 1 088 nm. The sea ice density exhibits a good linear relationship with the SIDS. Thus, we chose the 1 088 nm band to estimate $\mu_{650-690}$. $\mu_{650-690}$ can be expressed as a linear function of the reflection peak height (RPH) of sea ice at 1 088 nm:

$$\mu_{650-690} = A \cdot H_{rp, 1 088} + B, \quad (10)$$

where $H_{rp, 1 088}$ is the reflection peak height of sea ice at 1 088 nm; A and B are coefficients. $H_{rp, 1 088}$ can be calculated as follows:

$$H_{rp, 1 088} = R_{1 088} - \hat{R}_{1 088}, \quad (11)$$

where $R_{1 088}$ is the reflectance of sea ice at 1 088 nm; and $\hat{R}_{1 088}$ is the interpolation reflectance obtained based on the reflectance at 1 028 nm and the reflectance at 1 220 nm. As it is difficult to measure $\mu_{650-690}$ directly in the field, we estimated $\mu_{650-690}$ using Eq. (8c) based on the reflectance in Fig. 2 and the sea ice thickness in Table 1. Figure 8 illustrates the scatter diagram of $\mu_{650-690}$ vs. $H_{rp, 1 088}$. As shown in Fig. 8, $\mu_{650-690}$ exhibits a good linear relationship with $H_{rp, 1 088}$, including a highly correlated fitting effect (the coefficient of determination is 0.985). The regression analysis suggests that $A=0.271 9$ and $B=0.005 2$.

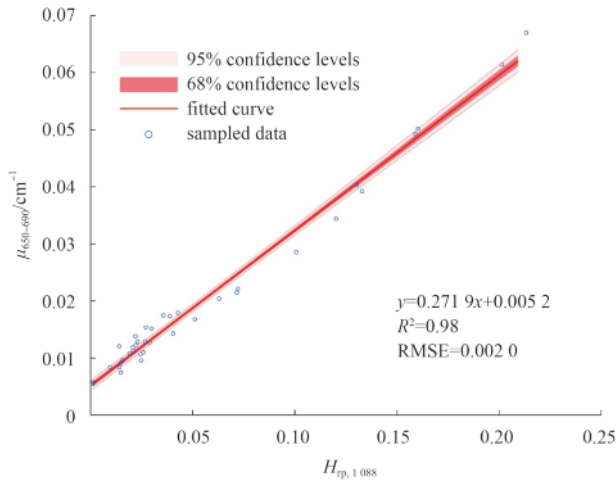


Fig. 8. Scatter diagram of $\mu_{650-690}$ vs. $H_{rp, 1\,088}$. $\mu_{650-690}$ is the integrated attenuation coefficient of sea ice for the 650–690 nm band. $H_{rp, 1\,088}$ is the reflection peak height of sea ice at 1 088 nm.

4 Data processing

In this section, the atmospheric correction of Hyperion data was performed firstly. Then the sea ice extent was extracted from the corrected Hyperion data. At last, the sea ice thicknesses was estimated by the SEMSIT and the sea ice thickness was also estimated from the albedo generated by the corrected Hyperion data for comparison.

4.1 Atmospheric correction of Hyperion data

Because the atmospheric absorption of the solar shortwave radiation decreases the radiation energy that reaches the optical remote sensing satellite sensor and atmospheric scattering of the solar shortwave radiation introduces the atmospheric scattering energy in the viewing direction of the sensor, atmospheric correction is required to remove the effects of the atmospheric absorption and scattering and accurately acquire the earth's surface reflectance. In this paper, the fast line-of-sight atmospheric analysis of spectral hypercubes (FLAASH) atmospheric correction module (Anderson et al., 2002) in the environment for visualizing images (ENVI) remote sensing image processing software was used to perform an atmospheric correction with the Hyperion data and to obtain the sea ice reflectance. The primary atmospheric parameters used in the FLAASH are listed in Table 3. The values of the parameters were inferred from meteorological data observed at the nearest meteorological station in the area covered by the Hyperion data (Fig. 4).

4.2 Extraction of sea ice from Hyperion data

The acquired images also cover partial land areas, it is necessary to remove those land areas from the acquired images. Because the reflectance values of seawater and sea ice are relatively small within the shortwave infrared (SWIR) band (Xie et al.,

2006), the threshold division method is adopted to distinguish sea areas from land areas. In this paper, the iterative self-organizing data analysis technologies algorithm (ISODATA) was used to classify Hyperion images into open water and sea ice classes. The ISODATA is a commonly used classification method for hyperspectral images. In this classification method, the class of each pixel can be dynamically adjusted based on the initial input parameters, and the classes can be “merged” and “divided” to obtain the optimal classification results (Kauufman and Rousseeuw, 2009; Tong et al., 2006). Generally, it is inevitable that there are some errors associated with sea ice extraction, such as errors caused by the ice in lakes or rivers, snow on land, sun glitter, and so on. Therefore, to reduce the effect of the errors on subsequent processing, we corrected the errors through visual interpretation after the automatic computer-based classification. The extraction results of sea ice is shown in Fig. 9.

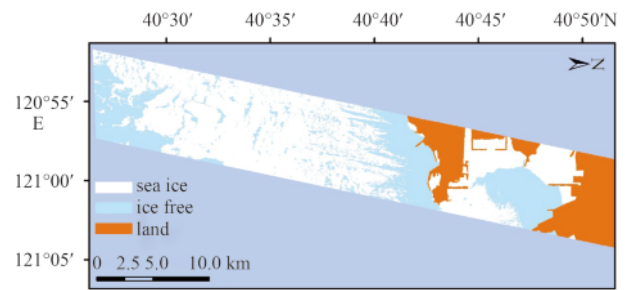


Fig. 9. Sea ice extraction results (acquired on January 21, 2008).

4.3 Estimation of sea ice thickness with SEMSIT

Owing to radiometric calibration and atmospheric correction errors, the obtained remote sensing data were not completely consistent with the reflection peak position of the *in situ* measured reflectance; however, the spectral shapes were similar. Therefore, to adapt Eq. (8b) based on the characteristics of Hyperion remote sensing data during sea ice thickness inversion, the methods for calculating the parameters in Eq. (8b) can be expressed as follows:

$$R_{1\,088} = \max(R_{b85}, \dots, R_{b114}), \quad (12)$$

$$\hat{R}_{1\,088} = 0.6875 \times \min(R_{b85}, \dots, R_{b94}) + 0.3125 \times \min(R_{b106}, \dots, R_{b114}), \quad (13)$$

$$R_{650-690} = \text{mean}(R_{b30}, \dots, R_{b34}), \quad (14)$$

$$R_{1\,550-1\,750} = \text{mean}(R_{b140}, \dots, R_{b160}), \quad (15)$$

where $\max(R_{bm}, \dots, R_{bn})$ is the maximum reflectance from Band m to Band n ; $\min(R_{bm}, \dots, R_{bn})$ is the minimum reflectance from Band m to Band n ; and $\text{mean}(R_{bm}, \dots, R_{bn})$ is the average reflectance from Band m to Band n . The parameters calculated using the above methods were substituted into Eq. (8b). Equation (8b) can now be combined with the parameterization scheme to calculate the sea ice thickness.

4.4 Estimation of sea ice thickness from albedo

According to Xie et al., (2003), Yuan et al. (2012), and Su and Wang (2012), the EM assumes that the relationship between the sea ice thickness and albedo is exponential. The EM can be expressed as follows:

Table 3. Hyperion image radiometric calibration status

Parameter	Value
Atmospheric model	mid-latitude winter
Water column multiplier	1.3
Aerosol model	maritime
Initial visibility	40 km
CO ₂ mixing ratio	3.90×10^{-4}

$$\alpha(h) = \alpha_{\max} [1 - \kappa \exp(-\mu_a h)], \quad (16)$$

where h is the sea ice thickness; $\kappa = 1 - \alpha_{\text{sea}}/\alpha_{\max}$, where $\alpha_{\max}$ is the reflectance corresponding to an infinite ice thickness and is equal to 0.7 and α_{sea} is the albedo of sea water with a value of 0.1; μ_a is the attenuation coefficient, which is set to 1.209 according to Xie et al. (2003) and Su and Wang (2012); and $\alpha(h)$ is the sea ice albedo corresponding to a certain sea ice thickness. According to Xie et al. (2003) and Yuan et al. (2012), $\alpha(h)$ can be calculated as follows:

$$\alpha(h) = 0.423 \times \text{mean}(R_{b23}, \dots, R_{b33}) + 0.577 \times \text{mean}(R_{b37}, \dots, R_{b57}, R_{b79}, R_{b83}, \dots, R_{b96}), \quad (17)$$

where $\text{mean}(R_{b23}, \dots, R_{b33})$ and $\text{mean}(R_{b37}, \dots, R_{b57}, R_{b79}, R_{b83}, \dots, R_{b96})$ correspond to Band 1 (580–680 nm) and Band 2 (725–1 100 nm) in the AVHRR data, respectively.

5 Results and analysis

In this section, the first validation was performed. The sea ice thicknesses estimated by the SEMSIT and the EM are compared with the forecasted sea ice thickness provided by the NMEFC. The effect of spatial resolution on the sea ice thickness estimation error is also discussed through comparing Hyperion data with the MODIS data.

5.1 Sea ice thickness retrieval with the SEMSIT and EM models

Figures 10 and 11 represent the ice thickness maps obtained based on estimations using the SEMSIT and the EM, respectively. The sea ice thicknesses estimated with the SEMSIT ranged from 2 cm to 38 cm. The sea ice thicknesses estimated with the EM ranged from 4 cm to 92 cm. The sea ice thickness distribution shapes estimated with the two models are nearly the same. However, the sea ice thicknesses estimated using the EM are larger than those estimated using the SEMSIT. The distribution of land-fast ice thickness estimated with the SEMSIT is quite different from that estimated with the EM. This is particularly obvious in areas close to the coast, where the sea ice thicknesses estimated with the EM are above 50 cm, which is nearly two times larger than the thicknesses estimated with SEMSIT. As shown in Fig. 12, for both land-fast sea ice and drift sea ice, the frequency histogram trends of the sea ice thickness variations estimated by the SEMSIT are also different from those estimated by EM. Their differences mainly lie in the average and standard deviation. The average and standard deviation of the SEMSIT estimation are all smaller than those associated with the EM estimation for both land-fast sea ice and drift sea ice.

Table 4 shows a comparison between the forecasted and es-

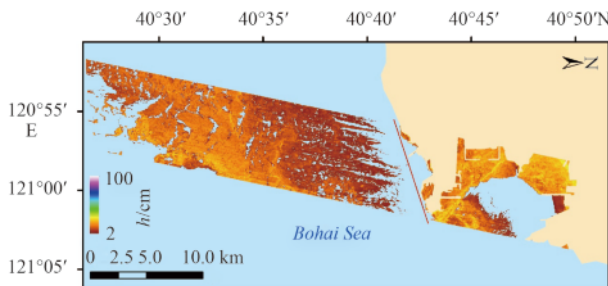


Fig. 10. Sea ice thickness estimated using SEMSIT. To the north (right) of the red line is land-fast sea ice, and to the south (left) is drift sea ice.

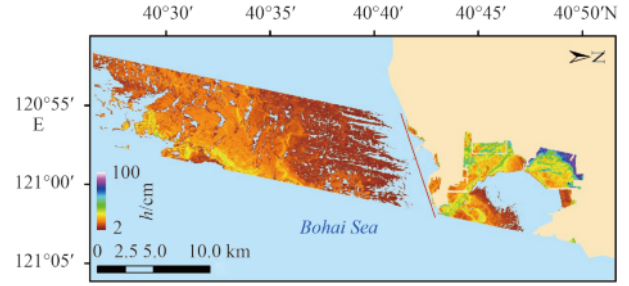


Fig. 11. Sea ice thickness estimated using EM. To the north (right) of the red line is land-fast sea ice, and to the south (left) is drift sea ice.

timated general sea ice thicknesses. The forecasted sea ice thickness is provided by the NMEFC. The general thickness was calculated as the average plus the standard deviation for the SEMSIT and EM estimations. The extents of land-fast sea ice and drift sea ice are indicated in Figs 10 and 11 for the SEMSIT and the EM, respectively. Table 4 shows that the relative differences associated with the SEMSIT estimation are 6.0% and 20.0% for land-fast sea ice and drift sea ice, respectively, and the relative differences of the EM estimation are 50.0% and 30.0% for land-fast sea ice and drift sea ice, respectively. The average relative difference is 13.0% for the SEMSIT estimation and 40.0% for the EM estimation. From the comparison above, it can be concluded that the relative difference and average relative difference of the SEMSIT estimation are much lower than those of the EM estimation. The main reasons for the accuracy improvement are that the integrated attenuation coefficient was estimated at the pixel level and the surface reflection correction was performed in the SEMSIT. In the EM, only one integrated attenuation coefficient was adopted for the pixels in the entire image.

5.2 The effect of spatial resolution on sea ice thickness estimation error

The results of Yuan et al. (2012) and Su and Wang (2012) suggested that the sea ice thickness estimated using the EM exhibited no significant error because the spatial resolutions of the adopted NOAA and MODIS data were relatively low (1.1 km and 500 m). Therefore, the majority of sea ice pixels were actually mixed ice-water pixels (Fig. 13) whose reflectance values were lower than those of pure sea ice. However, during the estimation of the sea ice thickness, the entire pixel was considered pure sea ice. Therefore, the impacts of mixed pixels were not considered. It is difficult to estimate the sea ice thickness, as well as the area ratio of sea ice in the pixel, using visible spectral data or thermal infrared data. Therefore, the sea ice thicknesses estimated using the AVHRR images and the MODIS images were not large. In the Hyperion images, the spatial resolution has increased by a factor of 30 or 16, causing the number of mixed pixels to significantly decline. The integrated attenuation coefficients of many pixels were undervalued, and the corresponding reflectance rapidly increased. Therefore, the estimated error associated with the sea ice thickness also rapidly increased using the EM.

6 Discussion and conclusions

Using the SEMSIT, the integrated attenuation coefficient and surface reflectance were estimated at the pixel level, eliminating the estimation error associated with the sea ice thickness caused by temporal and spatial heterogeneities in the integrated attenuation coefficient and sea ice surface reflectance. It is suitable for

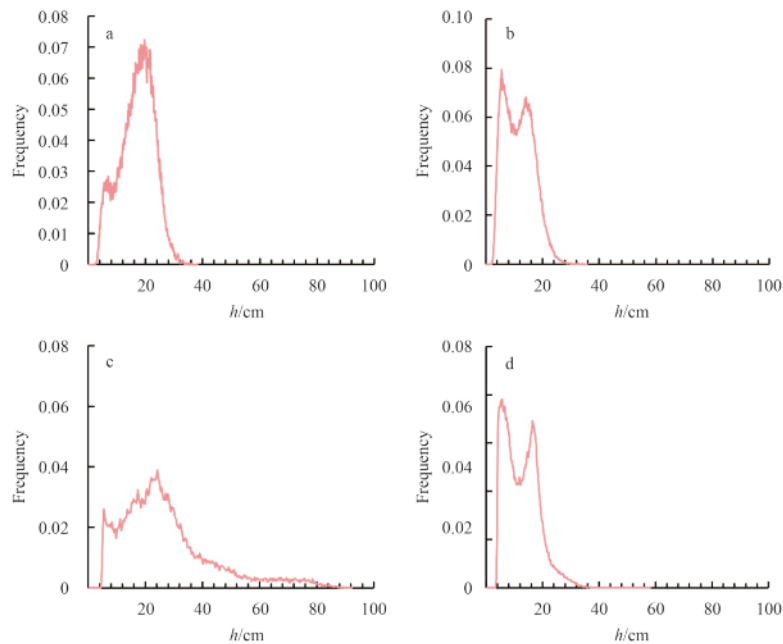


Fig. 12. Frequency histograms of sea ice thickness. a. Land-fast sea ice thickness estimated with SEMSIT, b. drift sea ice thickness estimated with SEMSIT, c. land-fast sea ice thickness estimated with EM, and d. drift sea ice thickness estimated with EM.

Table 4. Comparison of the forecasted and estimated general sea ice thicknesses

		Land-fast ice/cm	Drift ice/cm	Average/cm
General thickness	NMEFC forecast	10–25	5–15	–
	SEMSIT estimation	11–23	7–17	–
	EM estimation	11–43	7–19	–
Average thickness	NMEFC forecast	18	10	14
	SEMSIT estimation	17	12	15
	EM estimation	27	13	20
Relative difference compared NMEFC	SEMSIT estimation	6.0%	20.0%	13.0%
	EM estimation	50.0%	30.0%	40.0%

Note: The forecasted sea ice thickness is provided by the NMEFC. The general thickness is calculated as “average plus or minus standard deviation” for the SEMSIT and EM estimations.

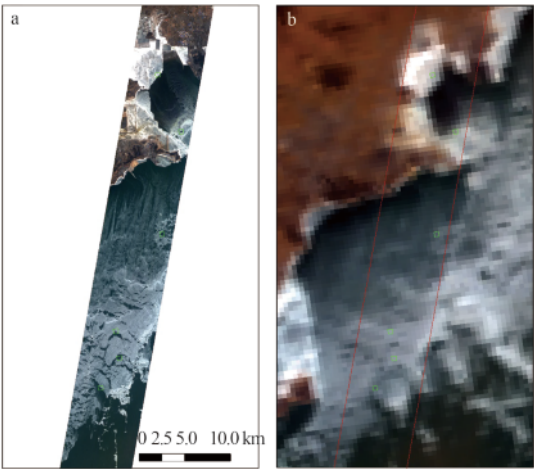


Fig. 13. Hyperion images (a) and MODIS images (b) (acquired at 10:30 on January 21, 2008). The size of the green box is 500 m×500 m, which has the same spatial resolution as the MODIS images. Pixels in the MODIS coverage area corresponding to pixels in the Hyperion image were previously considered sea ice, but they are actually mixed ice-water pixels.

estimating the thickness of first-year sea ice. The input parameters of the SEMSIT were based on the sea ice reflectance. Thus, the reflectance inversion accuracy directly affected sea ice inversion accuracy. For airborne and spaceborne hyperspectral data, both sensor-based spectral calibration and radiometric calibration cause errors. Spectral calibration led to deviations in the positions of peaks and valleys, and radiometric calibration affected the reflectance. Moreover, radiation is affected by atmospheric absorption and scattering during the imaging process. Therefore, the accuracies associated with the spectral calibration, radiometric calibration, and atmospheric correction of hyperspectral images should be enhanced as much as possible when estimating the sea ice thickness with the SEMSIT.

Because the number of existing hyperspectral sensors in orbit is limited, we acquired only one hyperspectral image containing ice and without cloud cover. Unfortunately, a synchronous sea ice thickness measurement was not performed that day. Therefore, it is necessary to obtain additional hyperspectral images to validate and improve the SEMSIT. In the future years, additional spaceborne hyperspectral remote sensing satellites will be launched and operated. For example, the China National Space Administration will launch a satellite (gao fen-5, GF-5) in 2015 that will be equipped with a wide-field hyperspectral imager

with a nanoscale spectral resolution of 400–2 500 nm.

We proposed a semi-empirical model of the sea ice thickness based on our analysis of the optical radiative transfer process in sea ice. Through practical application experiments, our findings suggest that the proposed model and the corresponding parameterization scheme can effectively overcome the sea ice thickness estimation errors caused by the temporal and spatial heterogeneities in the integrated attenuation coefficient and surface reflectance of sea ice. The study potentially provides a practical semi-empirical model and a feasible parameterization scheme that may be feasible for the sea ice thickness estimation using the hyperspectral data. In the future studies, we will carry out more synchronous measurement experiments to validate the proposed model and make it more practical and accurate.

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